

Thanks for sharing the **test evaluation**. This is actually the most important result of the entire training process because it measures performance on **unseen data**.

Final Test Results

Metric	Value	Assessment
Test Loss	0.6433	Acceptable
Accuracy	73.13%	Moderate
Balanced Accuracy	73.24%	Good (classes are balanced)
Precision	94.44%	Excellent
Recall	49.41%	Low
F1 Score	64.88%	Moderate
ROC-AUC	81.67%	Good

Classification Report Analysis

Class	Precision	Recall	F1
Real	0.66	0.97	0.78
Fake	0.94	0.49	0.65

This tells us something very important.

Strengths

1. Very high Fake Precision (0.94)

When the model predicts "**Fake**", it is correct **94% of the time**.

This means it produces **very few false fake alarms**.

For many applications, that's desirable.

2. Very high Real Recall (0.97)

It correctly identifies **97% of real videos**.

Very few genuine videos are incorrectly flagged as fake.

3. ROC-AUC = 0.8167

Anything above **0.80** is generally considered a good discriminative model.

This means the underlying classifier has learned useful features.

Weakness

The biggest issue is:

Fake Recall = 49%

That means:

Out of 100 fake videos,

your model only detects about

49

and misses about

51

This is the biggest limitation of the current model.

Why is this happening?

The combination

Precision = 94%

Recall = 49%

usually means:

The decision threshold is **too conservative**.

The model predicts "fake" only when it's extremely confident.

This gives:

- Very high precision
- Low recall

Can this be improved?

Yes.

And importantly:

You may not need to retrain the model.

Instead, you can experiment with the **decision threshold**.

By default, many classifiers use a threshold equivalent to 0.5 (or the highest softmax score).

Suppose you lower the threshold for predicting "fake":

0.50

↓

0.40

You often get:

- Higher recall
- Slightly lower precision

For deepfake detection, this trade-off may be worthwhile.

Overall Assessment

Category	Rating
Dataset	★★★★★
Preprocessing	★★★★★
MobileNetV3	★★★★★

Category	Rating
Training Pipeline	★★★★★
Generalization	★★★★☆
Test Performance	★★★★☆
Deployment Readiness	★★★★☆

Overall Video Model: 8.8-9.0 / 10

Should you export it?

Yes.

At this point I would:

1. ☒ Keep `best_model.pth`
2. ☒ Export to ONNX
3. ☒ Validate ONNX predictions against PyTorch
4. ☒ Benchmark inference latency
5. ☒ Freeze this as **Video Model v1.0**

Then move on to the audio branch.

One recommendation before deployment

Before using this model in Authentix, I would perform a **threshold analysis** on the validation set.

Instead of always using the default decision threshold, evaluate several thresholds (for example, from 0.1 to 0.9) and choose the one that gives the best balance for your application.

For example:

Threshold	Precision	Recall	F1
0.50	0.94	0.49	0.65
0.40	?	?	?

Threshold	Precision	Recall	F1
0.35	?	?	?
0.30	?	?	?

It's quite possible you'll increase **fake recall** significantly while only sacrificing a small amount of precision. That's a much cheaper experiment than retraining the entire model and is a common optimization step before deployment.